

AIM

TA.050 SYSTEM AVAILABILITY STRATEGY

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<Subject>

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Contents

Document Control	ii
Introduction	1
Purpose	1
Scope.....	1
Definitions	1
Critical Systems Availability	3
Unplanned System Outages	4
Planned System Outages	5
Physical (Hardware or Network) Component Failure	6
Database Server Failure	6
Application Server Failure.....	7
Network Failure	7
Data Center Failure	8
Client Desktop Failure	8
Software Component Failure	10
Database Software Failure	10
Application Software Failure	10
Century Date Failure	12
Century Date Compliance	12
Interface Failure	12
Reporting Failure	13
Maintenance Outages.....	14
Database Maintenance	14
Software Maintenance.....	14
Open and Closed Issues for this Deliverable	15
Open Issues.....	15
Closed Issues	15

Introduction

Purpose

The purpose of the **System Availability Strategy** document is to describe the high-level system strategies and approaches that <Company Short Name> will use to provide the level of system availability needed to support the new business operations.

System availability is a key element of an architecture design. The applications, databases, hardware, and networks must be selected or designed carefully so that integrated system maintains minimum continuous availability, and when outages do occur, they can be rectified within the allowable downtime metrics.

There are many points of failure in a complex system of applications, databases, Hardware, and networks. Only by properly considering all of them can the business be confident that unplanned outages will not create unacceptable operational downtime, especially during periods of peak demand such as the end of fiscal periods.



Note: It is important to remember that systems outages can often be avoided by a well-planned approach to systems management using proactive monitoring and maintenance wherever possible.

Scope

This document covers the following data centers:

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-
-

The following architecture components (database servers, file servers, and so on) are covered in this document:

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-
-

Definitions

Mean Time To Failure (MTTF)

Mean Time To Recover (MTTR)

Critical Systems Availability

This component summarizes the major (and most critical) system availability requirements.

Critical System Component	Component Type : System Application or Application Function	Availability Requirement	Acceptable Downtime	Comments

Unplanned System Outages

This section summarizes the main unplanned system outages due to failures in different components of the overall system and the strategy that will be used to handle to each. This table summarizes special architecture components or procedures that will need to be built into the architecture to handle system failure and provide the desired level of system availability.

Code	Unplanned Outage	Est. Outage Frequency without Strategy (1/MTTF)	Availability /Recovery Strategy	Est. Outage Frequency with Strategy (1/MTTF)	Predicted TTR	Comments
<Enter the codes you use below>	Single database server disk outage		2-way disk mirroring			Compute from MTTF per disk as per manufacturers spec. MTTF takes into account entire disk array
	Single database I/O controller outage					Compute from MTTF per disk as per manufacturers spec.
	Single database server CPU outage					
	Database server outage					
	Single Application Client File server disk outage					
	Single Application Client File server CPU outage					
	Application Client File server outage					
	Local area network outage					
	Wide area network outage within US					
	Wide area network outage outside US					

Planned System Outages

This section summarizes planned outages that are predicted to be needed for regular or routine system maintenance.

Code	Planned Outage	Outage Frequency	Predicted Schedule	Predicted Duration	Details
	Oracle Database Software Upgrade				
	Oracle Database Software Patch				
	Oracle Applications Client Software Upgrade				
	Oracle Applications Server Software Upgrade				
	Oracle Applications Software Patch				
	Oracle Database Cold Backup				

Physical (Hardware or Network) Component Failure

Each of the hardware or network components in the information systems architecture can be subject to failure in different ways and it is important to document how the different components will be protected against the various failures (if at all). This section focuses on physical component failures and the failure analysis is subdivided into failures for:

- database servers
- application file servers
- networks
- data centers
- client desktop machines

For each type of component, all possible physical failure events are considered for every machine or component with the scope of this analysis.

Database Server Failure

This section lists the causes, results and the strategy for providing continuing availability through database server failure or for recovering from the failure.

Failure Cause	Failure Code	Result	Availability/ Recovery Strategy	Comments
Loss of power supply	DS-1	Loss of all server resident application and database processing	Uninterrupted power supply	
Failed CPU	DS-2			
Failed System Bus	DS-3			
Failed Memory	DS-4			
Loss or corruption of a single disk	DS-5	Failure of database instances that access data or control structures on the disk	2-way disk mirror	
	DS-6	Failure of application processing for processes that need to read application code from the failed disk		
Loss of a disk I/O Controller	DS-7	Failure of database instances that access data or control structures through that controller	2-way disk mirror	
	DS-8	Failure of application processing for processes that need to read application code through the controller		

<Failure Cause>

Application Server Failure

This section lists the causes, results, and the strategy for providing continuing availability through application server failure, or for recovering from the failure. This applies to the following application servers:

- <Application Server Name>

Failure Cause	Failure Code	Result	Availability/ Recovery Strategy	Comments
Loss of power supply	FS-1	Loss of all application network client file services to desktop PCs	Uninterrupted power supply	
Failed CPU	FS-2	Loss of all application network client file services to desktop PCs		
Failed System Bus	FS-3	Loss of all application network client file services to desktop PCs		
Failed Memory	FS-4	Loss of all application network client file services to desktop PCs		
Loss or corruption of a single disk	FS-5	Loss of all application network client file services to desktop PCs		
Loss of a disk I/O Controller	FS-6	Loss of all application network client file services to desktop PCs		

<Failure Cause>

Network Failure

Network failures can occur within both the local and the wide area networks (LANs and WANs).

Network failures can affect business applications in two main areas:

- loss of communication between a client and a server machine, preventing user access to system in the Oracle Applications client-server architecture
- inability to transfer interface data between applications or databases network linked through the failed network segment

The network segment failure may also affect systems management capability by preventing:

- network transfer of files
- network monitoring of systems and applications

Failure Cause	Failure Code	Result	Availability/ Recovery Strategy	Comments
Loss of a LAN segment		Loss of communication between desktop clients and DB server. Application processing stops.		
Loss of Router				
Loss of WAN connection between US and AsiaPac		Loss of application interface data transfers.	Redundant leased line connection between US and AsiaPac data centers	

<Failure Cause>

Data Center Failure

Data Center failures entail the loss of functioning of all physical architecture components that are resident in the data center or the data center service area. The loss of a data center is an extreme or catastrophic event that is very rare but needs to be considered as a remote possibility.

Failure Cause	Failure Code	Result	Availability/ Recovery Strategy	Comments
Civil strife, earthquake, flood	DC-1	Loss of all application and database processing resident in the data center	Geographically remote disaster recovery site in...	

<Failure Cause>

Client Desktop Failure

Client desktop PCs can fail because of:

- network or disk copy virus infection
- non-standard (non IS approved) software loaded by users causing unpredictable PC resource usage
- disk corruption
- PC operating system bugs causing software failure.

Failure Cause	Failure Code	Result	Availability/ Recovery Strategy	Comments
Virus infection	PC-1	Partial or complete loss of regular PC functions	Backup spare PCs	Assume PC will require disinfecting and reload of standard IS approved software
Load of non-approved software	PC-2	Partial loss of regular PC function	IS standards imposed for PC usage by users. Backup spare PC can be made available	Deinstall of offending software. Reconfiguration of PC may be necessary.
Disk corruption	PC-3	Partial or complete loss of regular PC functions	PC disk backups. Backup spare PC can be made available	

<Failure Cause>

Software Component Failure

This section focuses on software component failures and the failure analysis is subdivided into failures at the level of:

- database software
- application software

This section does not include failures that are caused by the failure of physical system components. These failures are discussed in the Physical (Hardware or Network) Component Failure section.

Software component failures include these sources:

- user errors
- operations (maintenance) staff errors
- software bugs
- lack of adequate database space management

Database Software Failure

Failure Cause	Failure Code	Result	Availability/ Recovery Strategy	Comments
Database object dropped by IS Operations staff	DB-1	Users unable to access or process the dropped object	Restore using database backups	
Inadequate database space in structures	DB-2	Users unable to create new data in the affected database structure	Regular scheduled database monitoring. Emergency shutdown and reconfiguration of database.	

Application Software Failure

Failure Cause	Failure Code	Result	Availability/ Recovery Strategy	Comments
Database object dropped by IS Operations staff	DB-1	Users unable to access or process the dropped object	Restore using database backups	

Failure Cause	Failure Code	Result	Availability/ Recovery Strategy	Comments
Inadequate database space in structures	DB-2	Users unable to create new data in the affected database structure	Regular scheduled database monitoring. Emergency shutdown and reconfiguration of database.	

Century Date Failure

This section focuses on Century Date related failures. The failure analysis is subdivided into failures at the level of:

- interfaces
- reporting and analysis

This section does not include failures that are caused by the failure of physical system components. These failures are discussed in the Physical (Hardware or Network) Component Failure section.

Century Date failures include these sources:

- user errors
- operations (maintenance) staff errors
- software bugs (interfaces)
- improper access to data for reporting and analysis

Century Date Compliance

In the past, two character date coding was an acceptable convention due to perceived costs associated with the additional disk and memory storage requirements of full four character date encoding. As the year 2000 approached, it became evident that a full four character coding scheme was more appropriate.

In the context of the Application Implementation Method (AIM), the convention Century Date or C/Date support rather than Year2000 or Y2K support is used. Coding for any future Century Date is now the modern business and technical convention.

Every applications implementation team needs to consider the impact of the century date on their implementation project. As part of the implementation effort, all customizations, legacy data conversions, custom interfaces, data extraction mechanisms and architecture components need to be reviewed for Century Date compliance.

Interface Failure

Failure Cause	Failure Code	Result	Availability/ Recovery Strategy	Comments
Interface Truncates four character date	CD-1	Possible generation of date with wrong century	Repair interface, review and repair any data that was imported.	

Reporting Failure

Failure Cause	Failure Code	Result	Availability/ Recovery Strategy	Comments
Invalid selection of data in query due to invalid date logic in query.	CD-10	Report/Query may yield incorrect results.	Repair query, rerun report/query and validate results.	

Maintenance Outages

Maintenance outages are planned outage events that are required to perform some form of maintenance on the system. Examples of system maintenance events include:

- Regular database maintenance
 - database backups
 - database tuning
 - database space management
- Software maintenance
 - bug patches
 - software upgrade

Database Maintenance

Maintenance Event	Code	Availability During Maintenance	Maintenance Strategy	Comments
Cold database backup	DM-1	Database and applications unavailable	Use hot backups as much as possible. Cold backups once per week.	
Data archive				
Data purge				

Software Maintenance

Maintenance Event	Code	Availability During Maintenance	Maintenance Strategy	Comments
Oracle Database Software Upgrade	SM-1	Database and applications unavailable		
Oracle Database Software Patch	SM-2	Database and applications unavailable		
Oracle Applications Client Software Upgrade	SM-3	Applications unavailable		
Oracle Applications File Server Software Upgrade	SM-4	Applications unavailable	Minimize applications downtime by using Oracle Client Software Manager	
Oracle Applications Software Patch	SM-5	Applications unavailable	Minimize applications downtime by using Oracle Client Software Manager	

Open and Closed Issues for this Deliverable

Open Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date

Closed Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date